

Game Design Document: Stellar Reach

Genre: Sci-Fi UI-Based Trading & Management Simulator **Platform:** Mobile (Portrait Mode, One-Handed Play) **Pacing:** Turn-based / Action-Point based (No stressful real-time timers, play at your own pace)

1. The Core Philosophy (Why is it fun?)

Based on player psychology, "spread-sheet" games become boring when they lack narrative context and variable rewards. *Stellar Reach* fixes this by combining economic strategy with human elements.

- **Variable Ratio Reinforcement:** Market prices and travel events are semi-predictable but contain elements of chance (like a D&D dice roll), keeping dopamine levels engaged.
- **Emergent Narrative:** You aren't just trading "Iron" for "Credits." You are trading "Smuggled Med-Stims" while your pilot is dealing with space-sickness and your negotiator is having an existential crisis.
- **Cognitive Load Management:** The UI is clean and minimalist. The player only makes 2-3 meaningful choices per "turn," preventing decision paralysis and overwhelm.

2. Core Gameplay Loop

1. **Dock & Assess:** Arrive at a space station. Read the local news (which affects market prices).
2. **Trade & Upgrade:** Buy low, sell high. Spend profits on ship upgrades (like upgrading your hyper-drive battery—think of it as souping up a space-faring Chevy Bolt!) or crew therapy.
3. **Plot Course:** Choose a route to the next system. Safe routes take longer and consume more supplies; dangerous routes are fast but risk pirate encounters or cosmic anomalies.
4. **Transit Events:** Resolve 1-2 text-based events during travel using your crew's stats.

3. The UI Design (Clean & Immersive)

The entire game looks like a sleek, futuristic datapad. Dark mode by default, with neon accents (amber, cyan, violet).

- **Main Navigation (Bottom Tab Bar):**
 - **Market:** Dynamic graphs, buy/sell sliders.
 - **Ship:** A top-down blueprint of your vessel where you slot in upgrades.
 - **Crew:** Portraits of your crew members, showing their "Energy States" and stress levels.
 - **Helm:** The star map for choosing your next destination.

4. Key Mechanics

A. The "Energy & Psychology" Crew System (The RimWorld Element)

Your crew members aren't just stats; they are energy beings with psychological needs.

- **Energy States:** Instead of simple "HP," crew members have energy frequencies (e.g., "Radiant," "Grounded," "Chaotic," "Depleted").
- **Traits:** A crew member might have the "Empath" trait (gets better deals in trade, but absorbs stress from the rest of the crew) or the "Anxious Attachment" trait (works harder when paired with the Captain, but loses efficiency if left alone in the engine room).
- **Management:** You have to manage their mental health. If a crew member gets overstimulated by a pirate attack, you might need to bench them in the ship's "Sensory Deprivation Pod" to recover their energy.

B. Event Resolution (The D&D Element)

When traveling, you get pop-up scenarios (e.g., "You detect a distress beacon from a derelict freighter").

- You choose how to respond based on your crew.
- Sending your "Diplomat" (basically a Sci-Fi Bard) to hail them might result in a peaceful trade.
- Sending your "Engineer" to hack them might yield hidden cargo.
- Success is determined by background "dice rolls" modified by your crew's current Energy State and skills.

C. Dynamic Economy

Prices aren't static. A solar flare in the Apollo Sector might destroy their agricultural domes, causing the price of "Bio-Rations" to skyrocket. If you pay attention to the galactic news feed, you can anticipate these changes and make a fortune.

5. Progression & Open-Ended Goals

There is no "Game Over" unless your ship is destroyed and you go bankrupt.

- **Early Game:** Scraping by, taking small jobs, keeping your starter crew happy.
- **Mid Game:** Upgrading to a larger vessel, hiring specialized crew, manipulating local station economies.
- **Late Game:** Buying your own space stations, funding colony ships, or uncovering the mysteries of ancient cosmic entities.